

Group Project - Software Architecture

The total mark allocated: 30

Note:

- This is a **group** project. A group should have a **minimum of 3** members and must not exceed more than 6(six) members.
- All the work should be committed to a git repository regularly. **DO NOT WAIT UNTIL THE LAST MOMENT TO COMMIT YOUR WORK.**
- A viva session will be conducted via zoom and participation is mandatory. The date of the viva session will be communicated later.
- Divide the work among group members before starting the work.
- Copied assignments will be given zero marks.
- You can submit the git repository link as the deliverable of this project.
- Better to use one single repository to push all the code.
- All the commits should be merged to the master/main branch before the evaluation.

Evaluation Criteria

The individual contribution will be the main evaluation criterion. Other than that, the marks will be given by evaluating the following points.

- The completeness of the functionalities.
- Code quality, maintainability and extensibility.

Task

User Requirement:

Aligning with the Sri Lankan government's national policy of strengthening public services through digitalization, the Sri Lanka Police Department has identified the need to modernize traffic fine payments across the country. This initiative aims to eliminate inefficiencies in the traditional traffic fine settlement process and reduce the inconvenience faced by motorists. To achieve this, the department plans to introduce both a mobile application and a web portal for traffic fine collection.

When a driver is stopped for a violation, a traffic police officer issues a traffic fine sheet containing a unique fine reference number and a traffic fine category identifier. Drivers can then use the official mobile application to make an on-the-spot payment by entering these details.

If the fine is not paid immediately, drivers can later complete the payment using the same reference number and category identifier via the dedicated web portal by providing the required payment information. Once the payment is successfully processed, an SMS notification is sent to the traffic police officer, enabling the driver to retrieve their license.

Additionally, an administrative web portal should be implemented to allow senior officials to monitor traffic fine collections nationwide. This portal will provide insights such as district-wise total collections and breakdowns by fine categories, supporting better oversight and decision-making.

Assume that your team is assigned to build this system. You are required to build a separate backend Rest API application and separate front-end applications for each portal and an Android mobile application.

Minimum requirements:

- Android App to pay the traffic fines on-the-spot.
- Single page web application to pay the traffic fine online
- Admin web portal to monitor the traffic fine collection.
- Send an SMS to traffic police officers upon payment confirmation

Technologies to use:

- Use any language or framework to develop the backend REST API. (Java/Springboot preferred)
- Use any database technology as you wish. (Try to use JPA to access database)
- Use any framework to develop the web applications and mobile app as you wish. Ex: React, Vue, Angular, Flutter
- The authentication should be handled by using JWT tokens. You can use the Spring Security library for this.